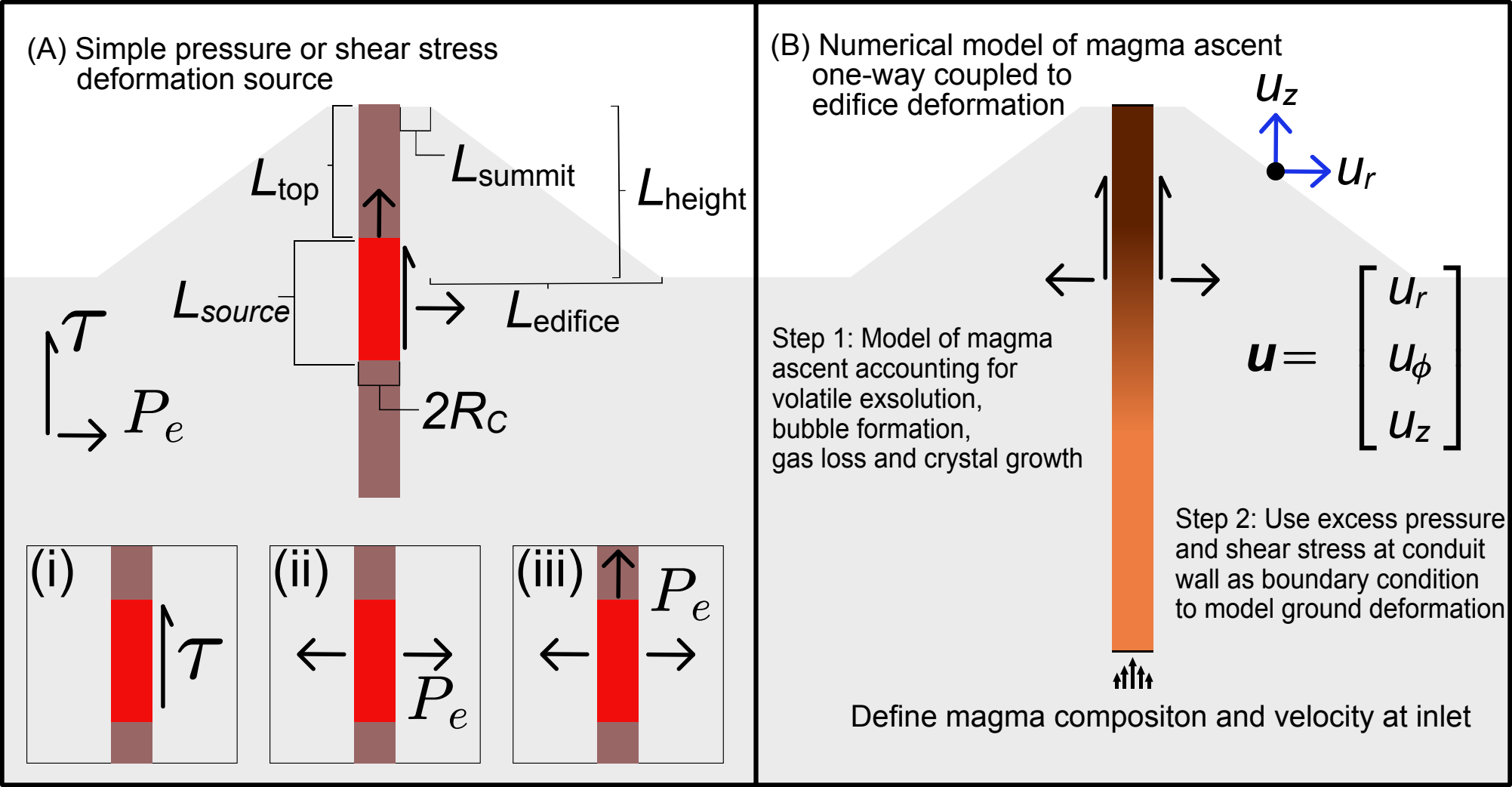
(A) Simple pressure or shear stress deformation source
L_{top}
L_{summit}
L_{height}
L_{source}
$L_{edifice}$
$2R_C$
τ
P_e
(i)
τ
(ii)
P_e
(iii)
P_e
(B) Numerical model of magma ascent one-way coupled to edifice deformation
u_z
u_r
Step 1: Model of magma ascent accounting for volatile exsolution, bubble formation, gas loss and crystal growth
$\boldsymbol{u} = \begin{bmatrix} u_r \\ u_\phi \\ u_z \end{bmatrix}$
Step 2: Use excess pressure and shear stress at conduit wall as boundary condition to model ground deformation
Define magma compositon and velocity at inlet